

R&D COOPERATION BETWEEN IMPATIENT RIVALS

Zhiyong Yao* and Bingyong Zheng†

*School of Management, Fudan University, Shanghai, China, and †School of Economics, Shanghai University of Finance and Economics, Shanghai, China

ABSTRACT

We analyse R&D cooperation between product-market competitors within a repeated-game framework with imperfect monitoring. When firms are patient enough, R&D cooperation is attainable without product-market collusion. However, if firms are less patient, we show that collusion in the product market is necessary to sustain R&D cooperation. Moreover, consumers can be better off when collusion is allowed in this case.

Keywords: collusion, imperfect monitoring, R&D cooperation

JEL classification numbers: C72, D82, L24

I. INTRODUCTION

Despite well-known benefits, such as internalizing spillover, reaping synergies, avoiding duplications, and sharing risks or costs, R&D cooperation suffers from moral hazard problems. While cooperation is mutually beneficial, once research joint ventures (RJVs) are formed, each involved firm has an incentive to free ride on the others, by contributing less or concealing its own know-how. The literature has examined formal contracts that can overcome the incentive problems (e.g., Bhattacharya *et al.*, 1992). But in many situations effective formal contracts are simply not practical (Levin, 2003). For example, when the R&D outcome is observable but not verifiable in court, no legally enforceable contract can be used. Under these circumstances, a long-term relationship is essential to provide firms with an incentive to cooperate. Many RJVs are set up for more than one project and have indeterminate ending dates.

It is well-known that in order to support cooperation in the long term, participating firms should be patient. However, firms could be less patient due to a variety of reasons. For instance, firms may have a high cost of capital and thus a high rate of time preference, or there may exist severe agency problems within firms that lead to myopic behaviours (Harrington, 1989). In these

Correspondence: Zhiyong Yao, School of Management, Fudan University, Room 236, Siyuan Building, 670 Guoshun Road, Shanghai 200433, China. Tel: 86-212-501-1066; Fax: 86-216-564-2411; Email: yzy@fudan.edu.cn. Conversations with Hongbin Cai, Martin Cripps, and Hong Feng were very helpful. We also thank seminar participants at Beijing University, Fudan University, and SHUFE. This research is supported by China National Natural Science Foundation (71273063), China Ministry of Education Humanities and Social Sciences Program (12YJC790236) and, Fudan Financial Research Center (2012FDFRCGD16). However, all errors are ours.

cases, sustaining R&D cooperation may require that firms establish a long-term relationship not only in the RJV, but also in the product market.

In this paper, we study R&D cooperation between two product-market competitors in an infinitely repeated game framework. Since firms' R&D resources, for example firm-specific know-how or human capital, are complements, it is mutually beneficial for them to form an RJV to develop cost-reducing technologies together.

Formally we can treat every period as consisting of two stages. In the first stage, the two firms simultaneously decide whether or not to exert effort in the RJV: to 'work' or to 'shirk'. In the second stage, after observing the outcome of the RJV, they engage in quantity or price competition in the product market. While the outcome of the RJV is commonly observable to both firms, the firms' effort choices are solely known to themselves. Because of uncertainties involved in the R&D process, the outcomes of the RJV can only imperfectly reflect the participating firms' effort choices.

We break our analysis into two parts. In the first part, we study R&D cooperation between patient firms. In the second part, we analyse R&D cooperation between impatient firms.

In the first part, we show that when firms are patient enough, R&D cooperation between them is possible without product-market collusion. We first specify the cooperative strategy in the spirit of Green and Porter (1984), which combines cooperation phases and punishment phases. We then find the threshold discount factor to support the cooperative equilibrium. Since punishments occur on the equilibrium path, the expected payoff is necessarily bounded away from the first-best. But we show that when the observation error is very small, the equilibrium payoff is arbitrarily close to the first-best, despite imperfect monitoring.

In the second part, we show that when firms are less patient, product-market collusion is necessary to sustain R&D cooperation. In other words, the product-market collusion facilitates R&D cooperation since the cooperation now is feasible at a lower discount factor. The intuition is as follows. The stake effect is generated by pooling the incentives for both R&D cooperation and the product-market collusion. Cheating can now be punished more severely by stopping both R&D cooperation and the product-market collusion, which decreases the threshold discount factor for supporting the R&D cooperation.

Furthermore, we demonstrate that consumers can be better off when firms are allowed to collude in the product market in this context. Product-market collusion usually reduces the quantity supplied and increases the price. This is the negative collusion effect. But collusion makes the R&D cooperation possible, which in turn saves product costs. This is the positive cost-saving effect. In our model, the cost-saving effect from successful R&D cooperation dominates the collusion effect. Therefore, consumers are better off even under product-market collusion.

Conventional wisdom holds that R&D cooperation may facilitate product-market collusion. Here we argue the opposite instead. We show that product-market collusion facilitates R&D cooperation when firms are not patient enough. Furthermore, R&D cooperation is under imperfect monitoring while product-market collusion is under perfect monitoring in our model. Hence, failure in R&D may destabilize otherwise stable collusion in the product market, which again, goes against conventional wisdom.

Therefore, we have made two contributions to the literature. First, we show how repeated interaction can sustain R&D cooperation between self-interested firms when effort choices are unobservable. Second, and more importantly, we demonstrate that allowing product market collusion can facilitate otherwise unsustainable R&D cooperation between firms that are not patient enough. This may sometimes increase consumer welfare, as well as firm profits. As far as we know, we are the first to make this finding.

R&D cooperation between rivals has attracted a great deal of attention from both economics and management science (see the survey papers by Veugelers, 1998; Caloghirou *et al.*, 2003).

Most existing literature on RJVs assumes a static framework with two-stage or three-stage games (D'Aspremont and Jacquemin, 1988; Kamien *et al.*, 1992; Suzumura, 1992; Choi, 1993, and many others). There are several papers studying R&D cooperation between rivals with repeated games. For example, Kesteloot and Veugelers (1995) show that the stability of cooperation is determined by the nature and magnitude of spillovers, relative to the nature and degree of product market competition. Eaton and Eswaran (1997) focus on technology trading within coalitions and product-market competition across coalitions. However, in contrast to our imperfect monitoring model, these papers deal with perfect monitoring situations. Our paper is also related to Bernheim and Whinston (1990), who show that firms may find it easier to collude if they interact in multiple markets in which they have asymmetric positions than if they interact in a single market. However, their paper also investigates the perfect monitoring case.¹

There are several papers discussing the linkage between R&D cooperation and collusion. Fershtman and Kamien (1992) show that cross-licensing of complementary technologies may facilitate collusion. Eswaran (1994) assumes that firms sell differentiated commodities and cross-licensing can enhance collusion by introducing the threat of increased rivalry for each firm's product. Using a continuous-time model, Martin (1995) assumes a stochastic R&D environment and collusion is ended with a discovery. He demonstrates that cooperation in process innovation enhances implicit collusion. Lambertini *et al.* (2002) focus on the interplay between firms' decisions in product development and the ensuing product market under horizontal or vertical differentiation. But in their model, R&D is deterministic. Miyagiwa (2009) considers the linkage between pre-discovery and post-discovery strategies and demonstrates that RJVs can eliminate post-discovery asymmetries and strengthen the firms' ability to collude. Our paper also discusses the relationship between R&D cooperation and product-market collusion, but argues that collusion facilitates R&D cooperation when firms are impatient.

There has been proliferating empirical evidence on RJVs since the mid-1980s (Veugelers, 1998). Many RJVs have indeterminate end dates once set up, which fits the repeated game framework used in this paper. For example, a very famous research consortium on information technology, SEMATECH, initiated in 1987, has finished numerous R&D projects and still involves many original partners such as Intel, AMD, HP, IBM, and Texas Instruments. Numerous case studies and econometric papers have shown that firms form RJVs in order to access complementary resources, and exploit research synergies as we have assumed in this paper (see Caloghirou *et al.*, 2003 for a survey).

The rest of the paper is organized as follows. In Section II, we describe the basic model. In Section III, we analyse the case when firms cooperate in the RJV while competing in the product market. In Section IV, we discuss the case when firms cooperate in both the RJV and the product market. In Section V, we extend our model to price competition in the product market. Finally, in Section VI, we present our conclusion.

II. THE MODEL

Two risk-neutral firms, 1 and 2, set up a RJV in order to develop cost-saving technologies together. Each period they simultaneously decide whether to work for or to shirk from the RJV. We denote their action sets by $\{W, S\}$, where W denotes 'work' and S denotes "shirk". It costs c

¹Poddar and Saha (2010) study an infinitely repeated play of a two-stage game of product innovation and market competition, and show that cooperation in giving R&D efforts is more easily sustained when firms compete in the product market *à la* Cournot than *à la* Bertrand. Yao and Zheng (2014) show that cooperation in multiple RJVs can mitigate the likelihood of cooperation breakdowns and reduce the efficiency loss due to imperfect monitoring.

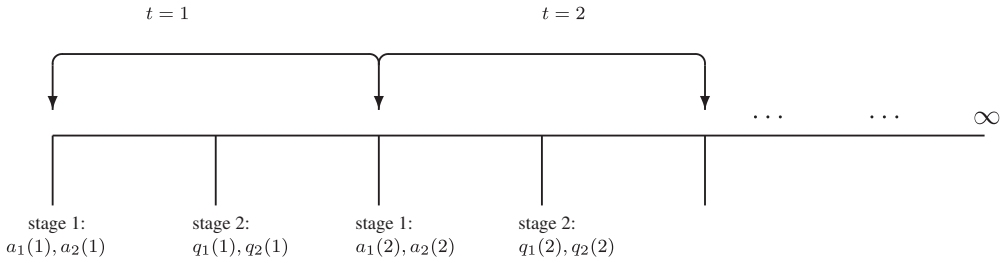


Fig. 1. Timing of events.

for each firm to choose W while there is no cost to choose S . The outcome of the RJV is either success or failure. If it is a success, the marginal cost is C^L at that period; if it is a failure, the marginal cost is C^H . For simplicity, we assume that any newly developed technology only works at the same period and then becomes useless from the next period on. Let r be the probability of success when the two firms choose $a(t) = (a_1(t), a_2(t)) \in \{W, S\} \times \{W, S\}$. Here we let

$$\begin{aligned} r^B &= Pr(C^L | W, W) \\ r^D &= Pr(C^L | W, S) = Pr(C^L | S, W) \\ 0 &= Pr(C^L | S, S) \end{aligned}$$

with $0 < 2r^D < r^B < 1$. The condition $r^B > 2r^D$ indicates that the combined efforts have greater effect on the probability of success than the sum of their individual efforts. It represents the synergy effect by combining the firms' complementary R&D efforts or resources, which is one of the main reasons why firms form RJVs. We further assume that the firm's behaviour, working or shirking, is observable by the other side, but both firms can observe the outcome of the RJV.

There are two stages in each period, the R&D stage followed by the product market interaction stage. We have just described the R&D stage. In the second stage both firms either compete or collude in the product market by choosing outputs $q(t) = (q_1(t), q_2(t))$, where $q_i(t) \in \mathbb{R}_+$ for $i = 1, 2$ and for all t . The demand function is $P = A - q_1 - q_2$ each period, where $q_1 + q_2 < A < 4C^H - 3C^L$. $q_1 + q_2 < A$ is purely a technical assumption while $A < 4C^H - 3C^L$ means the cost saving from successful R&D is large. The profit function for firm i is:

$$\pi_i(P, q_i) = \begin{cases} (P - C)q_i & \text{if } i \text{ shirks} \\ (P - C)q_i - c & \text{if } i \text{ works} \end{cases}$$

We summarize the timing of events in Figure 1.

III. COMPETITION IN THE PRODUCT MARKET

III.1 Period game

Before we discuss the repeated game, we first analyse the period game. In this section we assume that firms engage in Cournot competition in the product market.

There are three possible cases for the period game: both firms work; one works while the other shirks; and both firms shirk. Regardless of their effort choices in the R&D stage, firms engage in Cournot competition in the second stage. In the unique symmetric Cournot equilibrium, each

firm produces an output $q_i^L = (A - C^L)/3$ if the RJV succeeds and marginal cost equals C^L , and produces $q_i^H = (A - C^H)/3$ if the RJV does not succeed.

The expected payoffs of firm i for the above cases, are respectively:

$$E^B(\pi_i^*) = \frac{r^B(A - C^L)^2 + (1 - r^B)(A - C^H)^2}{9} - c \quad (1)$$

if both firms work;

$$E^W(\pi_i^*) = \frac{r^D(A - C^L)^2 + (1 - r^D)(A - C^H)^2}{9} - c \quad (2)$$

if firm i works while the other firm shirks;

$$E^S(\pi_i^*) = \frac{r^D(A - C^L)^2 + (1 - r^D)(A - C^H)^2}{9} \quad (3)$$

if firm i shirks while the other firm works;

$$E^N(\pi_i^*) = \frac{(A - C^H)^2}{9} \quad (4)$$

if both firms shirk, where $i = 1, 2$.

We make the following assumption:

Assumption 1.

$$r^B - r^D < \frac{9c}{[(A - C^L)^2 - (A - C^H)^2]} < \frac{r^B - r^D}{1 - r^D}.$$

The first inequality of Assumption 1 implies $E^S(\pi_i^*) > E^B(\pi_i^*)$ since $(r^B - r^D)[(A - C^L)^2 - (A - C^H)^2]/9 < c$. Given the assumption that $r^B > 2r^D$, it also ensures that $E^N(\pi_i^*) > E^W(\pi_i^*)$. The second inequality guarantees that $E^B(\pi_i^*) > E^N(\pi_i^*)$ and the discount factor for the cooperative equilibrium exists (see Lemma 1). Under Assumption 1, we can rank the expected payoffs as follows:

$$E^S(\pi_i^*) > E^B(\pi_i^*) > E^N(\pi_i^*) > E^W(\pi_i^*)$$

Hence, the period game between the two firms is a standard prisoners' dilemma game. The Nash equilibrium is for both firms to shirk.

Next we turn to the repeated game to see how the repeated interaction can help solve the prisoners' dilemma problem.

III.2 Repeated game

The repeated game $G^\infty(\delta)$ is an infinite repetition of the period game described above. Here δ is the common discount factor. At the end of the R&D stage for each period, the two firms observe a common signal $y \in \{C^L, C^H\}$. The information $h_i(t)$ observed by firm i at date t includes two parts: its own choice $a_i \in \{W, S\}$ and the public signal y .² A history for firm i at the end of period t is $h_i^t \equiv (h_i(1), \dots, h_i(t)) \in H_i^t$. And a public history $\bar{h}_i^t \in \bar{H}_i^t$ at the end of period t is $\bar{h}_i^t \equiv (y_1(1), \dots, y_i(t))$.

A strategy for firm i is the sequence $\sigma_i = (\sigma_i(1), \sigma_i(2), \dots)$, where each $\sigma_i(t)$ is a function mapping firm i 's history H_i^{t-1} to the action set $\{W, S\}^t$. At $t = 1$, the null history is mapped to the action set $\{W, S\}$. Firms maximize the long-run expected payoffs. We normalize each firm's

²Here we purposefully ignore the public signals in the production market since the two firms play Cournot competition in the product market anyway.

discounted sum of payoffs to the average period game payoffs.³ Given a strategy profile σ , firm i 's payoffs in the repeated game are

$$v_i(\sigma) = (1 - \delta) \left(\sum_{t=1}^{\infty} \delta^{t-1} \pi_i(a_i(t), a_j(t)) | \sigma \right)$$

A strategy profile σ constitutes a Nash equilibrium in the repeated game if and only if for firm $i = 1, 2$ and $j \neq i$,

$$v_i(\sigma) \geq v_i(\sigma'_i, \sigma_j) \quad \forall \sigma'_i$$

The equilibrium concept we use is a special class of Nash equilibrium called Perfect public equilibrium (PPE). A PPE is a profile of public strategies that constitutes a Nash equilibrium in the continuation game at any date t and given any public histories. It is recursive in the sense that it induces a PPE in every continuation game.⁴

III.3 Cooperative equilibrium

One equilibrium of the repeated game is for both firms to choose $a_i(t) = S$ for all t and for all history H_t^i . However, we consider a cooperative equilibrium in which both firms choose $a_i(t) = W$. Due to uncertainties with the innovation process, the cooperation will be unstable, as shown by Green and Porter (1984). We specify a *cooperative strategy* as follows. The game starts in a cooperation phase in which both firms choose W . They continue doing so as long as C^L is observed in the previous period. However, once the signal C^H is observed, the game enters into a punishment phase in which both firms choose S for T periods. After the punishment phase, a new cooperation phase starts again with both firms choosing W .

Proposition 1. *There exists a pair $(\underline{\delta}, T)$ such that for all $\delta \geq \underline{\delta}$, the cooperative strategy constitutes a PPE of the repeated game.*

Proof. Note that it is a best response for either firm to play S in a punishment phase, and neither firm has incentive to deviate. In a cooperation phase, firm i 's expected payoff from playing W is:

$$\begin{aligned} v_i = (1 - \delta) & \left[\frac{r^B(A - C^L)^2 + (1 - r^B)(A - C^H)^2}{9} - c \right] + \delta r^B v_i \\ & + \delta(1 - r^B) \left[(1 - \delta^T) \frac{(A - C^H)^2}{9} + \delta^T v_i \right] \end{aligned} \quad (5)$$

However, if firm i deviates by shirking at date t , assuming firm j follows the equilibrium strategy, firm i 's expected payoff would be:

$$\begin{aligned} v'_i = (1 - \delta) & \frac{r^D(A - C^L)^2 + (1 - r^D)(A - C^H)^2}{9} + \delta r^D v_i \\ & + \delta(1 - r^D) \left[(1 - \delta^T) \frac{(A - C^H)^2}{9} + \delta^T v_i \right] \end{aligned} \quad (6)$$

³The normalized payoffs v_i can be understood this way: if we spread the discounted sum of payoffs equally across all periods in the repeated game, then firm i would get v_i in every period of the repeated game.

⁴For detailed discussion of Perfect public equilibrium, see Fudenberg *et al.* (1994).

Hence, firm i 's incentive constraint for not shirking is:

$$v_i \geq v'_i$$

or

$$\begin{aligned} & (1 - \delta) \left[c - \frac{(r^B - r^D)[(A - C^L)^2 - (A - C^H)^2]}{9} \right] \\ & \leq \delta(r^B - r^D)v_i - \delta(r^B - r^D) \left[(1 - \delta^T) \frac{(A - C^H)^2}{9} + \delta^T v_i \right] \\ & = \delta(r^B - r^D)(1 - \delta^T) \left[v_i - \frac{(A - C^H)^2}{9} \right] \end{aligned} \quad (7)$$

Since $v_i > (A - C^H)^2/9$, the right-hand side of (C) is strictly positive, while the left-hand side goes to zero as $\delta \rightarrow 1$. We conclude that there exists $\underline{\delta} < 1$ and a related T such that for all $\delta \geq \underline{\delta}$, the cooperative strategy constitutes a perfect public equilibrium in the repeated game. ■

When firms engage in long-term R&D cooperation, the expected payoff from continuing cooperation in the future plays the role of 'carrot' that helps to discipline firms' behaviour. While shirking increases current payoff, it also increases the possibility of entering a punishment phase in which the firms' payoff will be lower every period. If firms are patient enough, shirking will not be a best response for the firm, as the expected loss in future payoff from discontinuing R&D cooperation exceeds the gain from shirking. In Proposition 1 we find such a threshold discount factor $\underline{\delta}$. If the common discount factor δ is larger than or equal to $\underline{\delta}$, both firms have no incentive to deviate from the cooperative strategy specified above.

We make two observations about the cooperative equilibrium characterized by $(\underline{\delta}, T)$. First, no firm has an incentive to deviate from the equilibrium strategy. Firms can figure out what the competing firm will do at any date, even if the choice of the competing firm is unobservable. From time to time, firms observe C^H when both firms are supposed to be working. Firms understand that this is purely due to bad luck with the innovation process, rather than unilateral deviation by the partner. Second, while understanding that no deviations ever took place in the previous period, firms nevertheless have incentives to revert to a punishment phase after observing C^H . Consequently, one would expect to observe non-cooperation from time to time. However, this does not imply firms' inability to cooperate on R&D. Instead, it is the threat of punishment that helps to ensure cooperation in the cooperation phase.

Nevertheless, reversion to the punishment phase is costly for both firms. Therefore, the expected payoff from the cooperative equilibrium strategy is necessarily bounded away from first-best outcome payoff. But if the observation error is small, the loss will be also small.

Proposition 2. For any $\varepsilon > 0$, if

$$r^B \geq \frac{E^S - E^B + r^D \varepsilon}{E^S - E^B + \varepsilon} \quad (8)$$

where E^S and E^B are defined respectively in (3) and (1), the cooperative equilibrium $(\underline{\delta}, T)$ yields a payoff $v_i \geq E^B - \varepsilon$ for firm i , $i = 1, 2$.

Proof. For any fixed $\delta \geq \underline{\delta}$, suppose that we can choose T such that (7) is just binding. In this case, we have:

$$(1 - \delta)(E^S - E^B) = \delta(r^B - r^D)[(1 - \delta^T)(v_i - E^N)]$$

or

$$\frac{(1 - \delta^T)(v_i - E^N)}{(1 - \delta)} = \frac{(E^S - E^B)}{\delta(r^B - r^D)} \quad (9)$$

Note that the equilibrium payoff can be simplified as

$$v_i = E^B - \frac{\delta(1 - r^B)[(1 - \delta^T)(v_i - E^N)]}{1 - \delta}$$

where E^N is defined in (4).

Plugging (9) into the equilibrium payoff gives

$$v_i = E^B - \frac{(E^S - E^B)(1 - r^B)}{(r^B - r^D)}$$

But condition (8) implies that

$$\frac{(E^S - E^B)(1 - r^B)}{(r^B - r^D)} \leq \varepsilon$$

Hence, we conclude that $v_i \geq E^B - \varepsilon$ under condition (8). ■

The intuition for this result is as follows. In repeated games with imperfect monitoring, the accuracy of the monitoring technology determines the efficiency loss due to observation errors. When the observation error is very small as indicated in (8), the equilibrium payoff is arbitrarily close to the first-best payoff E^B .

Next we will determine the minimum discount factor that is necessary for R&D cooperation to be feasible when firms engage in product-market competition. Note that T stands for the severity of punishment. The less patient firms are, the more severe the punishment needs to be to ensure no deviation. For our purpose, we only need to find the minimum discount factor $\underline{\delta}^*$ when the most severe punishment will be applied, $T = \infty$.

Lemma I. $\underline{\delta}^*$ exists under Assumption 1.

Proof. Let $T = \infty$ in order to obtain $\underline{\delta}^*$. In this case,

$$v_i = (1 - \delta) \left[\frac{r^B(A - C^L)^2 + (1 - r^B)(A - C^H)^2}{9} - c \right] + \delta r^B v_i + \delta(1 - r^B) \frac{(A - C^H)^2}{9}$$

When the discount factor equals $\underline{\delta}^*$, firm i is indifferent between choosing W and S . Hence, firm i 's incentive constraint for working is:

$$(1 - \underline{\delta}^*) \left[c - \frac{(r^B - r^D)[(A - C^L)^2 - (A - C^H)^2]}{9} \right] = \underline{\delta}^*(r^B - r^D) \left[v_i - \frac{(A - C^H)^2}{9} \right] \quad (10)$$

This condition implies that

$$\begin{aligned} v_i &= \frac{r^B(A - C^L)^2 + (1 - r^B)(A - C^H)^2}{9} - c - \frac{\underline{\delta}^*(1 - r^B)}{1 - \underline{\delta}^*} \left[v_i - \frac{(A - C^H)^2}{9} \right] \\ &= \frac{r^B(A - C^L)^2 + (1 - r^B)(A - C^H)^2}{9} - c - \frac{1 - r^B}{r^B - r^D} \\ &\quad \left[c - \frac{(r^B - r^D)[(A - C^L)^2 - (A - C^H)^2]}{9} \right] \end{aligned} \quad (11)$$

The equilibrium payoff can be further simplified as:

$$v_i = \frac{(A - C^L)^2}{9} - \frac{(1 - r^D)c}{r^B - r^D} \quad (12)$$

Plugging the payoff v_i into (10), we get the minimum discount factor as:

$$\underline{\delta}^* = \frac{9c - (r^B - r^D)[(A - C^L)^2 - (A - C^H)^2]}{9r^D c}$$

Hence, $0 < \underline{\delta}^* < 1$ exists under Assumption 1. ■

When the common discount factor for firms is $\delta \geq \underline{\delta}^*$, firms will cooperate on R&D with product-market competition. However, what if the common discount factor is smaller than the threshold discount factor, or $\delta < \underline{\delta}^*$? Will it still be possible for firms to collaborate in R&D? We examine this issue in the following section.

IV. COLLUSION IN THE PRODUCT MARKET

In the previous section we have assumed that firms can not collude in the product market. Under such a circumstance, when $\delta < \underline{\delta}^*$, no R&D cooperation is feasible. In this section, we show when collusion in the product market is allowed, R&D cooperation may be possible even for some $\delta < \underline{\delta}^*$. Hence, we will allow firms to choose between ‘collude’ and ‘not collude’ in the product market from now on. If they choose ‘not collude’, they engage in Cournot competition in the product market. As before, we first consider the period game. Then we turn to the repeated interaction.

IV.1 Period game

In the extensive-form period game, firms first choose either ‘ W ’ or ‘ S ’ for the RJV in the R&D stage. After observing the outcome of RJV, firms then choose between ‘collude’ and ‘not collude’ in the product market.

Lemma 2. The unique Nash equilibrium for the period game is that both firms choose ‘ S ’ in the R&D stage, and then choose ‘not collude’ in the product market.

Proof. We first characterize the expected payoffs for firm i in the period game, with or without the product market collusion. Our discussion is corresponding to the three possible cases for the RJV analysed in Section III.3. Since we have examined all the cases when the firms do not collude at all, we shall not repeat them here.

1. Both firms choose W for the RJV. If both firms collude in the product market, the expected payoff for firm i is:

$$E^{B,M}(\pi_i^*) = \frac{r^B(A - C^L)^2 + (1 - r^B)(A - C^H)^2}{8} - c$$

If firm i chooses ‘collude’ while firm j chooses ‘not collude’ in the product market, the expected payoffs for them are, respectively:

$$E^{B,C}(\pi_i^*) = \frac{3[r^B(A - C^L)^2 + (1 - r^B)(A - C^H)^2]}{32} - c$$

and

$$E^{B,D}(\pi_j^*) = \frac{9[r^B(A - C^L)^2 + (1 - r^B)(A - C^H)^2]}{64} - c$$

Given that both firms choose W for the RJV, it is a standard prisoners' dilemma game for the product market interaction stage since

$$E^{B,D}(\pi_i^*) > E^{B,M}(\pi_i^*) > E^B(\pi_i^*) > E^{B,C}(\pi_i^*)$$

where $E^B(\pi_i^*)$ is defined in (1). The unique Nash equilibrium is for both firms to choose 'not collude' in the product market.

2. Firm i chooses W and firm j chooses S for the RJV. If they collude in the product market, the expected payoffs for them are, respectively:

$$E^{W,M}(\pi_i^*) = \frac{r^D(A - C^L)^2 + (1 - r^D)(A - C^H)^2}{8} - c$$

and

$$E^{S,M}(\pi_j^*) = \frac{r^D(A - C^L)^2 + (1 - r^D)(A - C^H)^2}{8}$$

If firm i colludes while firm j does not collude in the product market, the expected payoffs for them are respectively:

$$E^{W,C}(\pi_i^*) = \frac{3[r^D(A - C^L)^2 + (1 - r^D)(A - C^H)^2]}{32} - c$$

and

$$E^{S,D}(\pi_j^*) = \frac{9[r^D(A - C^L)^2 + (1 - r^D)(A - C^H)^2]}{64}$$

On the other hand, if firm j colludes while firm i does not collude in the product market, the expected payoffs for them are respectively:

$$E^{W,D}(\pi_i^*) = \frac{9[r^D(A - C^L)^2 + (1 - r^D)(A - C^H)^2]}{64} - c$$

and

$$E^{S,C}(\pi_j^*) = \frac{3[r^D(A - C^L)^2 + (1 - r^D)(A - C^H)^2]}{32}$$

Given that firm i chooses W and firm j chooses S for the RJV, it is a standard prisoners' dilemma game for both i and j in the product market interaction stage since

$$E^{W,D}(\pi_i^*) > E^{W,M}(\pi_i^*) > E^W(\pi_i^*) > E^{W,C}(\pi_i^*)$$

and

$$E^{S,D}(\pi_j^*) > E^{S,M}(\pi_j^*) > E^S(\pi_j^*) > E^{S,C}(\pi_j^*)$$

where $E^W(\pi_i^*)$ and $E^S(\pi_j^*)$ are defined in (2) and (3), respectively. The Nash equilibrium is that both firms choose 'not collude' in the product market.

3. Both firms choose S for the RJV. If they collude in the product market, the expected payoff for either firm is:

$$E^{N,M}(\pi_i^*) = \frac{(A - C^H)^2}{8}$$

If firm i colludes while firm j chooses not to collude in the product market, the expected payoffs for them are, respectively:

$$E^{N,C}(\pi_i^*) = \frac{3(A - C^H)^2}{32}$$

and

$$E^{N,D}(\pi_j^*) = \frac{9(A - C^H)^2}{64}$$

Given that both firms choose S for the RJV, it is a standard prisoners' dilemma game for the product market, since

$$E^{N,D}(\pi_i^*) > E^{N,M}(\pi_i^*) > E^N(\pi_i^*) > E^{N,C}(\pi_i^*)$$

where $E^N(\pi_i^*)$ is defined in (4). The Nash equilibrium is both firms choosing 'not collude' in the product market.

In all three cases listed above, it is a dominant strategy for firm i to choose 'not collude' in the product market. Meanwhile, it is also a dominant strategy for firm i to play ' S ' for the RJV. Therefore the unique equilibrium for the period game is for both firms to choose ' S ' in the R&D stage, and then choose 'not collude' in the second stage. ■

IV.2 Cooperation on two fronts

We now examine the infinite repetition of the period game described in Section IV.1. We focus on a cooperative equilibrium in which both firms choose to cooperate in R&D and to collude in the product market. As before, we specify a collusive strategy that consists of both cooperation phases and punishment phases.

Consider the following strategy profile σ^m . The two firms start the game in a cooperation phase, wherein they both choose ' W ' in the R&D stage; then if the RJV succeeds, they both choose 'collude' in the product market. They continue doing so in the next period. However, if the RJV fails, they enter into a punishment phase, in which they engage in Cournot competition in the product market in the second stage, and from the next period on, they stop both R&D cooperation and product-market collusion for T periods. After T periods of punishment, they return to the cooperation phase again. In the case that the RJV succeeds in stage one, but one firm does not (or both firms do not) collude in the product market, the two firms stop R&D cooperation and product-market collusion forever.

While effort choices for the RJV are unobservable, cheating in the product market is perfectly observable by one's opponent. As long as no firm deviates from the collusive strategy in the product market, the most severe punishment as specified in the collusive strategy σ^m , terminating R&D cooperation and product market collusion, will not be triggered by any event occurring in the product market. Punishment will be triggered by failure in the RJV only on the equilibrium path when firms follow the collusive strategy.

We further make the following assumption:

Assumption 2.

$$r^B - r^D < \frac{9c}{[9(A - C^L)^2/8 - (A - C^H)^2]}^5$$

⁵If this condition is violated, the threat of not colluding in the product market alone is enough to ensure cooperation in R&D, which makes R&D cooperation easier to be sustained but renders the analysis here irrelevant.

Given the collusive strategy σ_i^m , firm i 's expected payoff when both firms follow the strategy is:

$$v_i^m = (1 - \delta) \left[\frac{r^B(A - C^L)^2}{8} + \frac{(1 - r^B)(A - C^H)^2}{9} - c \right] + \delta r^B v_i^m + \delta(1 - r^B) \left[(1 - \delta^T) \frac{(A - C^H)^2}{9} + \delta^T v_i^m \right] \quad (13)$$

Note that if the RJV succeeds after both firms choose ' W ', the two firms then collude in the product market. However, if the RJV fails after both firms choose to work, the two firms then compete in the product market. Therefore, the expected payoff for current period is $r^B(A - C^L)^2/8 + (1 - r^B)(A - C^H)^2/9 - c$ as in (13).

Firm i can deviate from the collusive strategy in three ways: (i) shirking in the RJV but colluding in the product market; (ii) working for the RJV but not colluding in the product market; and (iii) shirking in the RJV and not colluding in the product market. Conditional on its counterpart following the collusive strategy σ^m , firm i 's payoffs for the three kinds of deviations are respectively:

$$v_i^1 = (1 - \delta) \left[\frac{r^D(A - C^L)^2}{8} + \frac{(1 - r^D)(A - C^H)^2}{9} \right] + \delta r^D v_i^m + \delta(1 - r^D) \left[(1 - \delta^T) \frac{(A - C^H)^2}{9} + \delta^T v_i^m \right]$$

$$v_i^2 = (1 - \delta) \left[\frac{9r^B(A - C^L)^2}{64} + \frac{(1 - r^B)(A - C^H)^2}{9} - c \right] + \delta \frac{(A - C^H)^2}{9}$$

and

$$v_i^3 = (1 - \delta) \left[\frac{9r^D(A - C^L)^2}{64} + \frac{(1 - r^D)(A - C^H)^2}{9} \right] + \delta \frac{(A - C^H)^2}{9}$$

Then we proceed to prove the following result:

Proposition 3. *There exists a pair $(\underline{\delta}', T)$ such that for all $\delta \geq \underline{\delta}'$, the collusive strategy σ^m constitutes a PPE of the repeated game.*

Proof. Note that under Assumption 2,

$$v_i^3 - v_i^2 = (1 - \delta) \left[c - \frac{9(r^B - r^D)(A - C^L)^2}{64} + \frac{(r^B - r^D)(A - C^H)^2}{9} \right] \geq 0$$

This leaves us two incentive constraints to work with.

We now show that $v_i^m \geq v_i^1$ or

$$(1 - \delta) \left[c - (r^B - r^D) \left(\frac{(A - C^L)^2}{8} - \frac{(A - C^H)^2}{9} \right) \right] \leq \delta(r^B - r^D)(1 - \delta^T) \left[v_i^m - \frac{(A - C^H)^2}{9} \right] \quad (14)$$

Note that the right-hand side of equation (14) is strictly positive while the left-hand side goes to zero as δ goes to one. Hence, there exists $\underline{\delta}'$ and a related T such that for any $\delta \geq \underline{\delta}'$, $v_i^m \geq v_i^1$.

It happens that when $v_i^m \geq v_i^1$, we can obtain $v_i^1 - v_i^3 > 0$. To see this result, note that

$$\begin{aligned} v_i^1 - v_i^3 &> \delta r^D \left[v_i^m - \frac{(A - C^H)^2}{9} \right] - \frac{(1 - \delta)r^D(A - C^L)^2}{64} \\ &= \frac{r^D}{r^B - r^D} \left\{ \delta(r^B - r^D) \left[v_i^m - \frac{(A - C^H)^2}{9} \right] - \frac{(1 - \delta)(r^B - r^D)(A - C^L)^2}{64} \right\} \end{aligned}$$

As mentioned before, under Assumption 2,

$$c - \frac{9(r^B - r^D)(A - C^L)^2}{64} + \frac{(r^B - r^D)(A - C^H)^2}{9} \geq 0$$

It implies

$$\begin{aligned} \frac{(1 - \delta)(r^B - r^D)(A - C^L)^2}{64} &\leq (1 - \delta) \left[c - (r^B - r^D) \left(\frac{(A - C^L)^2}{8} - \frac{(A - C^H)^2}{9} \right) \right] \\ &< \delta(r^B - r^D) \left[v_i^m - \frac{(A - C^H)^2}{9} \right] \end{aligned}$$

The last inequality is because of (14).

Hence, we conclude that there exists $\underline{\delta}'$ and a related T such that for any $\delta \geq \underline{\delta}'$, firm i has no incentives to take any possible deviation, and the collusive strategy σ^m constitutes a PPE of the repeated game. $\square$

When firms can engage in long-term relationships in both R&D and the product market, the stake effect is generated. Now shirking can be punished more severely by stopping both the R&D cooperation and the product market collusion. The firms have more to lose, thus they have more incentive to maintain the cooperation, which decreases the threshold discount factor for supporting the R&D cooperation. Indeed, the threshold discount factor $\underline{\delta}$, specified in Proposition 1, is larger than the threshold discount factor $\underline{\delta}'$, specified in Proposition 3. For the same reason, the minimum threshold discount factor $\underline{\delta}^*$, specified in Lemma 1, is also larger than its counterpart when firms can cooperate in both fronts. This result is summarized in the following proposition. Again we let $T = \infty$ to obtain the minimum $\underline{\delta}'$, denoted by $\underline{\delta}^{**}$.

Proposition 4. For any fixed T , $\underline{\delta}' < \underline{\delta}$; When $T = \infty$, $\underline{\delta}^{**} < \underline{\delta}^*$.

Proof. Compare inequalities (7) and (14). When both inequalities are binding and fix T , we can obtain $\underline{\delta}' < \underline{\delta}$.

We let $T = \infty$,

$$\begin{aligned} v_i^m &= (1 - \delta) \left[\frac{r^B(A - C^L)^2}{8} + \frac{(1 - r^B)(A - C^H)^2}{9} - c \right] + \delta r^B v_i^m \\ &\quad + \delta(1 - r^B) \frac{(A - C^H)^2}{9} \end{aligned} \quad (15)$$

When the discount factor equals $\underline{\delta}^{**}$, firm i is indifferent between working and shirking,

$$\begin{aligned} (1 - \underline{\delta}^{**}) \left[c - (r^B - r^D) \left(\frac{(A - C^L)^2}{8} - \frac{(A - C^H)^2}{9} \right) \right] \\ = \underline{\delta}^{**}(r^B - r^D) \left[v_i^m - \frac{(A - C^H)^2}{9} \right] \end{aligned} \quad (16)$$

This condition implies that

$$v_i^m = \frac{r^B(A - C^L)^2}{8} + \frac{(1 - r^B)(A - C^H)^2}{9} - c - \frac{\underline{\delta}^{**}(1 - r^B)}{1 - \underline{\delta}^{**}}$$

$$\left[v_i^m - \frac{(A - C^H)^2}{9} \right] = \frac{(A - C^L)^2}{8} - \frac{(1 - r^D)c}{r^B - r^D} \quad (17)$$

Plugging the payoff v_i^m into (16), we get the minimum discount factor

$$\underline{\delta}^{**} = \frac{c - (r^B - r^D)[(A - C^L)^2/8 - (A - C^H)^2/9]}{r^D c}$$

Under Assumptions 1 and 2, $0 < \underline{\delta}^{**} < 1$.

In addition, we note that

$$\underline{\delta}^{**} = \frac{9c - (r^B - r^D)[(A - C^L)^2 - (A - C^H)^2]}{9r^D c} - \frac{(r^B - r^D)(A - C^L)^2}{72r^D c} < \underline{\delta}^*$$

■

The product-market collusion facilitates R&D cooperation in the sense that firms can cooperate at a lower threshold discount factor. When the common discount factor is δ such that $\underline{\delta}^{**} \leq \delta < \underline{\delta}^*$, R&D cooperation is not possible if the product-market collusion is prohibited. However, when the product market collusion is allowed, firms will still be able to cooperate in R&D, even if they are not patient enough to make cooperation in R&D alone feasible.

Whereas R&D cooperation reduces production costs, and thus, increases consumer surplus, as well as firms' profits, collusion in product market reduces consumer surplus and social welfare. The question naturally arises, what is the net effect on social welfare and consumer surplus? To answer this question, we examine consumer surplus with and without the product-market collusion when $\underline{\delta}^{**} \leq \delta < \underline{\delta}^*$.

*Proposition 5. Suppose $\underline{\delta}^{**} \leq \delta < \underline{\delta}^*$, consumer surplus is higher when firms can cooperate on both fronts than when they cannot.*

Proof. Without product market collusion and $\delta < \underline{\delta}^*$, firms can not engage in R&D cooperation. In this case, total output $Q = 2(A - C^H)/3$ and market price $P = (A + 2C^H)/3$ in the Cournot equilibrium. The per period consumer surplus is:

$$CS = \frac{2(A - C^H)^2}{9} \quad (18)$$

Now suppose firms can collude in the product market. In this case, firms will be able to establish long-term R&D cooperation as we showed previously. When firms cooperate on R&D but the research project fails, consumer surplus for the period is the same as the no R&D cooperation case, $CS_f = CS$. However, if the R&D succeeds, firms have a lower production cost but collude in the product market. In this case, total output $Q = (A - C^L)/2$, price $P = (A + C^L)/2$ and consumer surplus $CS_s = (A - C^L)^2/8$. This implies that the expected consumer surplus equals

$$CS^1 = \frac{r^B(A - C^L)^2}{8} + \frac{2(1 - r^B)(A - C^H)^2}{9} \quad (19)$$

for a period in the cooperation phase. Given $A < 4C^H - 3C^L$, $CS^1 > CS$. In addition, we already know that the consumer surplus for a period in the punishment phase equals CS .

Therefore, the overall consumer surplus is higher when firms can cooperate in both fronts than when there is no collusion, when $\underline{\delta}^{**} < \delta < \underline{\delta}^*$. Hence, we obtain Proposition 5. ■

The intuition for this result is simple. Without product-market collusion, cooperation in R&D is not sustainable when firms are not very patient, $\underline{\delta}^{**} < \delta < \underline{\delta}^*$. In this case, the production cost is high. Even if firms compete in the product market, equilibrium output will be low and market price will be high, because firms always have a high marginal cost. When the product market collusion is possible, cooperation in R&D becomes sustainable, and firms have a low production cost whenever the RJV succeeds. In this case, equilibrium output is actually higher, while price is lower, than the case when no R&D cooperation is possible.

Two forces of opposite directions are at work here. On the one hand, collusion tends to reduce the total output and increase the market price. On the other hand, successful R&D decreases the production cost, which leads to a higher output and a lower price. The positive effect of cost-saving is more than offsetting the negative effect from collusion. Hence, on average, firms produce higher total outputs and charge lower prices. Consumers are better off than the case of no collusion and no R&D cooperation. In other words, more value is created when product market collusion facilitates R&D cooperation. Furthermore, both the firms and the consumers capture higher surplus in this context.

V. PRICE COMPETITION IN THE PRODUCT MARKET

Our previous analysis focuses on the situation in which firms engage in quantity competition in the product market. One may therefore wonder whether a similar result holds when firms engage in price competition in the product market. To answer this question we consider a two-firm Bertrand oligopoly model, where firms produce differentiated products that are imperfect substitutes. The demand function for firm i 's ($i = 1, 2$) product is $q_i(P_i, P_j) = A - \alpha P_i + \beta P_j$ where $\alpha > \beta > 0$. Like before, the two firms can engage in R&D cooperation to reduce the constant marginal cost from C^H to C^L at cost c .

When the two firms have the same constant marginal cost C^s , $s \in \{H, L\}$, we can solve the model to get

$$P_1 = P_2 = \frac{A + \alpha C^s}{2\alpha - \beta}, \quad q_1 = q_2 = \frac{\alpha[A - (\alpha - \beta)c]}{2\alpha - \beta}$$

And the equilibrium profit for firm i is:

$$\pi_i(C^s) = \frac{\alpha[A - (\alpha - \beta)C^s]^2}{(2\alpha - \beta)^2}$$

The following two assumptions are similar to Assumption 1 in Section III and Assumption 2 in Section IV:

Assumption 1A.

$$r^B - r^D < \frac{(2\alpha - \beta)^2 c}{\alpha[A - (\alpha - \beta)C^L]^2 - \alpha[A - (\alpha - \beta)C^H]^2} < \frac{r^B - r^D}{1 - r^D}$$

Assumption 2A.

$$r^B - r^D < \frac{c}{[A - (\alpha - \beta)C^L]^2 / 4(\alpha - \beta) - \alpha[A - (\alpha - \beta)C^H]^2 / (2\alpha - \beta)^2}$$

We first consider a situation where firms can not collude in the product market. Under Assumption 1A, the R&D cooperation game between the two firms is a prisoner's dilemma game.

A Green–Porter style cooperative strategy constitutes an equilibrium in the repeated R&D game, provided that the two firms are patient enough. In this case, the expected equilibrium payoff v_i is:

$$v_i = \frac{\alpha[A - (\alpha - \beta)C^L]^2}{(2\alpha - \beta)^2} - \frac{(1 - r^D)c}{r^B - r^D}$$

It can be shown that the minimum discount factor $\underline{\delta}^{*f}$ exists given Assumption 1A, and

$$\underline{\delta}^{*f} = \frac{(2\alpha - \beta)^2 c - (r^B - r^D)\alpha[(A - (\alpha - \beta)C^L)^2 - (A - (\alpha - \beta)C^H)^2]}{(2\alpha - \beta)^2 r^D c}$$

Next, we show when collusion in the product market is allowed, R&D cooperation may be possible even for some $\delta < \underline{\delta}^{*f}$. As before, except for choosing ‘work’ or ‘shirk’ in R&D, firms can choose ‘collude’ or ‘not collude’ in the product market. We consider the same collusive strategy σ^m used in Section IV.2. Of course, if firms choose ‘not collude’ here, they engage in Bertrand competition rather than Cournot competition.

When the two firms collude in the product market, the collusive profit firm i obtains is:

$$\pi_c(C^s) = \frac{[A - (\alpha - \beta)C^s]^2}{4(\alpha - \beta)} > \pi_i(C^s), \quad s = H, L$$

If Assumption 2A is satisfied, another discount factor $\underline{\delta}^{**f}$ exists where

$$\underline{\delta}^{**f} = \underline{\delta}^{*f} - \frac{(r^B - r^D)\beta^2[A - (\alpha - \beta)C^L]^2}{4(\alpha - \beta)(2\alpha - \beta)^2 r^D c}$$

When firms are patient enough, $\delta > \underline{\delta}^{**f}$, the collusive strategy σ^m constitutes a perfect public equilibrium of the repeated game in which firms can cooperate on two fronts. Thus, while R&D cooperation is not possible in the absence of the product market collusion for some discount factor $\delta \in [\underline{\delta}^{**f}, \underline{\delta}^{*f})$, firms will be able to cooperate in R&D when product market collusion is allowed as long as $\delta \geq \underline{\delta}^{**f}$.

Therefore, we have shown that the main results obtained in previous sections remain when firms compete in price in the product market.

VI. CONCLUSION

We study a repeated game in which two firms confront each other in both the RJV and the product market every period. Cooperation in the RJV is desirable because of the high complementarity of R&D efforts between the product-market rivals. Within the imperfect monitoring setting, we first show that when firms are patient enough, R&D cooperation between them is possible without product-market collusion. We also demonstrate that when the observation error is very small, the equilibrium payoff is arbitrarily close to the first-best, despite imperfect monitoring. Then we show that when firms are less patient, product-market collusion is necessary to sustain R&D cooperation.

Conventional wisdom believes that R&D cooperation may facilitate product-market collusion, but this paper argues the opposite. We show that product-market collusion facilitates R&D cooperation when firms are not patient enough. Furthermore, we present that consumers can be better off when firms collude, because the positive cost-saving effect dominates the negative collusion effect.

Several interesting extensions of our model are possible. For example, if the two firms are asymmetric, such as having different R&D costs, different R&D capabilities, or different discount factors, how to sustain the cooperative equilibrium is worth further investigation.⁶ In this paper, we assume the product-market collusion is under perfect monitoring. However, if the product-market collusion is under imperfect monitoring (Green and Porter, 1984), how to sustain R&D cooperation and product-market collusion remains an open question.

REFERENCES

- Bernheim, D. and Whinston, M. (1990). 'Multimarket contact and collusive behavior', *Rand Journal of Economics*, 21, pp. 1–26.
- Bhattacharya, S., Glazer, J. and Sappington, D. (1992). 'Licensing and the sharing of knowledge in research joint ventures', *Journal of Economic Theory*, 56, pp. 43–69.
- Caloghirou, Y., Ioannides, S. and Vonortas, N. (2003). 'Research joint ventures', *Journal of Economic Survey*, 17, pp. 541–70.
- Choi, J. P. (1993). 'Cooperative R&D with product market competition', *International Journal of Industrial Organization*, 11, pp. 553–70.
- D'Aspremont, C. and Jacquemin, A. (1988). 'Cooperative and noncooperative R&D in duopoly with spillovers', *American Economic Review*, 78, pp. 1133–37.
- Eaton, C. and Eswaran, M. (1997). 'Technology-trading coalitions in supergames', *Rand Journal of Economics*, 28, pp. 135–49.
- Eswaran, M. (1994). 'Cross-licensing of competing patents as a facilitating device', *Canadian Journal of Economics*, 27, pp. 689–708.
- Fershtman, C. and Kamien, M. (1992). 'Cross licensing of complementary technologies', *International Journal of Industrial Organization*, 10, pp. 329–48.
- Fudenberg, D., Levine, D. and Maskin, E. (1994). 'The folk theorem with imperfect public information', *Econometrica*, 62, pp. 997–1039.
- Green, E. and Porter, R. (1984). 'Noncooperative collusion under imperfect price information', *Econometrica*, 52, pp. 87–100.
- Harrington, J. (1989). 'Collusion among asymmetric firms', *International Journal of Industrial Organization*, 7, pp. 289–307.
- Kamien, M., Muller, E. and Zang, I. (1992). 'Research joint venture and R&D cartels', *American Economic Review*, 82, pp. 1293–306.
- Kesteloot, K. and Veugelers, R. (1995). 'Stable R&D cooperation with spillovers', *Journal of Economics and Management Strategy*, 4, pp. 651–72.
- Lambertini, L., Podder, S. and Sasaki, D. (2002). 'Research joint ventures, product differentiation, and price collusion', *International Journal of Industrial Organization*, 20, pp. 829–54.
- Levin, J. (2003). 'Relational incentive contracts', *American Economic Review*, 93, pp. 835–57.
- Martin, S. (1995). 'R&D joint ventures and tacit product market collusion', *European Journal of Political Economy*, 11, pp. 731–41.
- Miyagiwa, K. (2009). 'Collusion and research joint ventures', *Journal of Industrial Economics*, 57, pp. 768–84.
- Poddar, S. and Saha, B. (2010). 'Product innovation and stability of collusion', *Economics Bulletin*, 30, pp. 1392–400.
- Suzumura, K. (1992). 'Cooperative and noncooperative R&D in an oligopoly with spillovers', *American Economic Review*, 82, pp. 1307–20.

⁶Yao (2011) discusses R&D cooperation between asymmetric firms using a repeated game framework and shows that the less capable firm obtains more from the RJV.

- Veugelers, R. (1998). 'Collaboration in R&D: an assessment of theoretical and empirical findings', *De Economist*, 146, pp. 419–33.
- Yao, Z. (2011). 'Bribe to cooperate', *Managerial and Decision Economics*, 32, pp. 535–44.
- Yao, Z. and Zheng, B. (2014). 'Feasibility, stability, and multiple research joint ventures', *Scottish Journal of Political Economy*, 61, pp. 196–210.